Training-Free Ranking from Pairwise Comparisons via Acyclic Graph Construction

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Abstract

Ranking from weighted pairwise comparisons asks for a global order that respects observed preferences as well as possible. We study a training-free, deterministic pipeline that builds an acyclic backbone on the comparison digraph using an MWFAS-inspired local-ratio heuristic, restores high-weight arcs under exact cycle-safe reinsertion, optionally applies a secondary weighted min-cut exchange, and extracts scores with optional Phase C refinement (adjacent-swap order updates followed by order-preserving ternary magnitude search). The local-ratio and exact add-back components follow prior lineage; this paper formalizes ranking–MWFAS optimum-value equivalence and practical guarantee boundaries, corrects a fixed-topological-position proxy weakening of add-back, and expands classical and GNN evaluation under timeout-aware common-completion analysis. Empirically, the canonical reachability-aware method (OURS-REACH) is strong on simple/naive upset metrics against several baselines, remains competitive rather than uniformly superior to SpringRank on family-aware checks, and is weaker than BTL and corrected RankCentrality on upset_ratio. It is slower than most lightweight classical estimators yet substantially faster than archived trained GNNRank runs under an end-to-end protocol; the near-complete Finance graph marks a large-dense scalability boundary.

Keywords: Ranking; Pairwise comparisons; Feedback arc set; Directed graphs; Training-free methods

1 Introduction

Ranking from pairwise comparisons arises throughout statistics, machine learning, and network analysis, from sports analytics to preference aggregation. Methods typically return a real-valued score vector whose induced order is evaluated by upset-based losses He et al. (2022). Classical estimators are fast and training-free; learning-based directed GNNs such as GNNRank can be accurate but introduce training cost, hyper-parameter sensitivity, and reduced reproducibility.

Directed comparison graphs almost always contain cycles arising from noise, incompleteness, and local inconsistencies. A combinatorial response is to delete a set of directed arcs so that the remaining graph is acyclic, then read a ranking from a topological order of the resulting directed acyclic graph (DAG). This is the Minimum Weighted Feedback Arc Set (MWFAS) viewpoint: minimize the total weight of deleted comparisons subject to acyclicity. Related spectral and synchronization methods already note connections between ranking inconsistency and feedback structure Fogel et al. (2014); Cucuringu (2015). MWFAS is NP-hard Karp (1972), so exact minimization is generally impractical, yet the formulation supplies a precise consistency-restoration objective for ranking.

This paper develops a practical, training-free, deterministic, and time-bounded ranking pipeline rather than a new general MWFAS approximation algorithm (§1.2). Prior work by Vahidi and Koutis (2025) already used the local-ratio heuristic of Demetrescu and Finocchi Demetrescu and Finocchi (2003) with weight-prioritized exact cycle-safe reinsertion and optional refinement. We clarify that lineage, correct an implementation-fidelity gap, add theory and a secondary structural repair, and expand empirical validation.

To make the ranking–MWFAS connection precise, we begin with formal definitions.

Definition 1 (Ranking Problem)**.** *Let $G = (V, E)$ be a directed graph, where V is the set of items and E is the set of directed pairwise-comparison edges. Each edge $(i, j) \in E$ has weight $w_{ij} > 0$, representing the strength of the preference for i over j. A* ranking *is a total order on V, represented equivalently by a bijection $R : V \to \{1, \ldots, |V|\}$, where smaller values indicate higher rank. An edge (i, j) is* inconsistent *with R if $R(i) > R(j)$. The goal is to find a ranking that minimizes the total weight of inconsistent edges:*

$$\min_{R} \sum_{(i,j)\in E:\ R(i)>R(j)} w_{ij}. \tag{1}$$

Definition 2 (Minimum Weighted Feedback Arc Set (MWFAS))**.** *Let $G = (V, E)$ be a directed graph with edge weights $w_{ij} > 0$ for each $(i, j) \in E$. A set of edges $F \subseteq E$ is a* feedback arc set *if removing F makes the graph acyclic, i.e., $(V, E \setminus F)$ is a directed acyclic graph (DAG). The MWFAS problem is*

$$\min_{F\subseteq E} \sum_{(i,j)\in F} w_{ij} \quad \text{s.t.} \quad (V, E \setminus F) \text{ is acyclic.} \tag{2}$$

The ranking objective (1) and MWFAS (2) share the same optimum *value*: the backward edges of any ranking form a feedback arc set of equal weight, and any feedback arc set induces a DAG whose topological orders have ranking disagreement no greater than the deleted weight, with equality for an inclusion-minimal (in particular, optimal positive-weight) feedback arc set. The correspondence between optimal rankings and optimal feedback arc sets is many-to-many rather than one-to-one; we treat this equivalence as a formal claim to be stated and proved, not merely asserted.

1.1 Scope, lineage, and contributions

The pipeline studied here (§2) builds an acyclic backbone with the local-ratio heuristic of Demetrescu and Finocchi (2003), reinserts high-weight arcs while preserving acyclicity, extracts scores from the resulting DAG, and optionally applies Phase C refinement (adjacent swaps that may change the order, then order-preserving ternary magnitude search). Several components are *prior*: local-ratio Phase A, exact cycle-safe add-back (including weight ordering), and ternary/refinement ideas already appear in Demetrescu and Finocchi (2003); Vahidi and Koutis (2025). What this paper contributes is therefore not a reinvention of that core, but the following.

1. **Formalization and guarantee boundaries.** Ranking–MWFAS optimum-value equivalence (Prop. 1), implementation complexity, and when the Demetrescu–Finocchi guarantee does and does not transfer (§2.9).
2. **Implementation-fidelity correction.** Restore exact reachability-aware reinsertion in place of a fixed-topological-position proxy (§2.5); ablations quantify the effect (§3.7).
3. **Secondary structural repair.** A conflict-triggered weighted min-cut exchange with monotone objective improvement (Prop. 3), with regime-specific empirics (§3.9).
4. **Expanded empirical validation.** Classical and GNN comparisons with timeout-aware common completions, paired statistics, family-aware sensitivity, and ablation/sensitivity analysis (§3). Dataset-count expansion is not treated as a primary novelty.

1.2 Related Work

1.2.1 Feedback Arc Set and MWFAS

The Minimum Weighted Feedback Arc Set (MWFAS) problem asks for a minimum-weight set of directed arcs whose removal makes a graph acyclic Karp (1972); Festa et al. (2020). Classical approximation results include divide-and-conquer methods based on spreading metrics and LP relaxations Even et al. (1995, 2000). Exact and parameterized algorithms target smaller or structurally restricted instances Baharev et al. (2021); Hecht et al. (2021); Raman and Saurabh (2006); Shea-Blymyer and Young (2021). Heuristic refinements emphasize minimal/stable feedback sets and initialization effects Cavallaro et al. (2024, 2025). Closely related is the maximum acyclic subgraph problem Hassin and Rubinstein (1994). We use MWFAS as a ranking consistency objective, not as a setting in which we claim a new general approximation algorithm.

1.2.2 Ranking from Pairwise Comparisons

Ranking from noisy and incomplete pairwise comparisons has been studied through statistical, spectral, random-walk, and optimization-based formulations Hochberg and Rabinovitch (2000); Chen et al. (2021); Shah and Wainwright (2016). Our experiments evaluate score-based outputs under the upset-style losses used in recent ranking benchmarks He et al. (2022), so we focus on the principal baseline families represented in that protocol.

1.2.3 Statistical and Spectral Ranking

BTL models pairwise win probabilities through latent abilities Bradley and Terry (1952). DavidScore aggregates unbalanced paired outcomes into per-item scores David (1987). Random-walk and spectral constructions include RankCentrality Negahban et al. (2017), SerialRank Fogel et al. (2014), SyncRank Cucuringu (2015), and SVD-based scores d'Aspremont et al. (2021). Graph scoring methods include eigenvector centrality Bonacich (1987), PageRank Page et al. (1999), and SpringRank De Bacco et al. (2018). Spectral and synchronization approaches already note connections between ranking inconsistency and feedback structure Fogel et al. (2014); Cucuringu (2015).

1.2.4 GNN-Based Ranking

Learning-based methods infer ranking scores from directed comparison graphs. GNNRank uses directed-GNN backbones with ranking-oriented losses He et al. (2022). We compare against the framework's DIGRAC He et al. (2021) and inception-style (ib) Tong et al. (2020) backbones. These methods can be accurate but incur training cost and hyperparameter sensitivity, which motivates a training-free alternative under fixed wall-clock budgets.

1.2.5 Local-Ratio Cycle Breaking, Add-Back Lineage, and Structural Repair

Demetrescu and Finocchi Demetrescu and Finocchi (2003) give a local-ratio cycle-breaking heuristic with an approximation guarantee parameterized by the length of a longest simple directed cycle, building on the local-ratio framework of Bar-Yehuda et al. (2004). Their Phase 2 reinserts deleted arcs subject to an *exact* cycle-safety test (equivalently: reinsert (u, v) iff v does not reach u in the current DAG), and they remark that ordering candidates by decreasing weight can improve solution quality. Exact cycle-safe reinsertion and weight-prioritized ordering are therefore prior; they are not newly invented here.

Component	DF03	VK25	This work
Local-ratio backbone	yes	yes	prior (used)
Exact cycle-safe reinsertion	yes	yes (spec.)	fidelity restore
Weight-ordered reinsertion	suggested	yes	prior (used)
Time-bounded practical impl.	–	partial	audited
Phase C refinement (swap + ternary)	–	ternary	prior (used)
Formal ranking↔MWFAS	–	asserted	proved
DF03 guarantee under timeout	N/A	not audited	qualified (no)
Weighted min-cut exchange	–	–	secondary new
Broad classical + GNN eval.	–	GNN-focused	expanded

Table 1: Novelty separation relative to Demetrescu–Finocchi (DF03) and Vahidi and Koutis (2025). "Fidelity restore" means restoring the established exact test after an implementation weakening, not inventing the test.

1.2.6 Relation to Vahidi and Koutis (2025)

Vahidi and Koutis (2025) operationalizes ranking via the Demetrescu and Finocchi (2003) local-ratio backbone with descending-weight exact cycle-safe reinsertion and order-preserving ternary refinement. A later practical codebase weakened reinsertion to a fixed topological-position proxy; multipass INS variants partially compensate and are ablation baselines only. Table 1 summarizes what is prior, restored for fidelity, or newly analyzed here.

2 Method and Theory

We describe a deterministic, training-free ranking pipeline for weighted pairwise-comparison graphs. The canonical pipeline is:

1. **Phase A (prior):** local-ratio-style cycle breaking Demetrescu and Finocchi (2003); Vahidi and Koutis (2025).
2. **Phase B (fidelity correction):** exact weight-prioritized reachability-aware reinsertion Demetrescu and Finocchi (2003); Vahidi and Koutis (2025).
3. **Optional secondary repair (new):** conflict-triggered weighted min-cut exchange.
4. **Phase C (prior lineage):** topological score extraction and optional refinement (adjacent-swap then order-preserving ternary) Vahidi and Koutis (2025).

Fixed-topological-position multipass variants (INS1–3) are retained only as ablation contrasts (§3.7).

2.1 Problem Setting

Let $V = \{1, \ldots, n\}$ be items and $G = (V, E, w)$ a directed graph with strictly positive arc weights $w : E \to \mathbb{R}_{>0}$. Antiparallel arcs are allowed. The solver input is a sparse weighted adjacency matrix converted to arrays $(\texttt{src}, \texttt{dst}, \texttt{w})$ of length $m = |E|$. The output is a score vector $s \in \mathbb{R}^n$ with larger values indicating higher rank. Formal ranking and MWFAS definitions appear in §1.

2.2 Ranking–MWFAS Optimum-Value Equivalence

Proposition 1 (Optimum-value equivalence)**.** *Let $G = (V, E, w)$ with $w > 0$. Write*

$$\mathrm{OPT}_{\mathrm{rank}} = \min_R \sum_{(i,j)\in E:\, R(i)>R(j)} w_{ij}, \qquad \mathrm{OPT}_{\mathrm{MWFAS}} = \min\bigl\{\textstyle\sum_{e\in F} w_e : F \subseteq E,\ (V, E \setminus F)\ \textit{acyclic}\bigr\}.$$

Then $\mathrm{OPT}_{\mathrm{rank}} = \mathrm{OPT}_{\mathrm{MWFAS}}$. The correspondence between optimal rankings and optimal feedback arc sets is many-to-many: we do not *claim a bijection between arbitrary rankings and arbitrary feedback arc sets.*

Proof. Write $L(R) = \sum_{(i,j)\in E:\, R(i)>R(j)} w_{ij}$ for the ranking loss of R.

($\mathrm{OPT}_{\mathrm{MWFAS}} \le \mathrm{OPT}_{\mathrm{rank}}$.) Fix any ranking R and let $F_R = \{(i,j) \in E : R(i) > R(j)\}$. Every retained arc is forward under R, so R is a topological order of $(V, E \setminus F_R)$ and F_R is a feedback arc set of weight $L(R)$. Hence $\mathrm{OPT}_{\mathrm{MWFAS}} \le L(R)$ for every R, and taking the best ranking yields $\mathrm{OPT}_{\mathrm{MWFAS}} \le \mathrm{OPT}_{\mathrm{rank}}$.

($\mathrm{OPT}_{\mathrm{rank}} \le \mathrm{OPT}_{\mathrm{MWFAS}}$.) Let $F^\star$ be an optimal feedback arc set and let R be any topological order of the DAG $(V, E \setminus F^\star)$. Every retained arc is forward under R, so the set of backward arcs of R is contained in $F^\star$. Therefore $L(R) \le w(F^\star) = \mathrm{OPT}_{\mathrm{MWFAS}}$, which implies $\mathrm{OPT}_{\mathrm{rank}} \le \mathrm{OPT}_{\mathrm{MWFAS}}$. □

Remark 1. *If F is inclusion-minimal, then for every topological order R of $(V, E \setminus F)$ one has $L(R) = w(F)$. Optimal feedback arc sets under strictly positive weights are inclusion-minimal, but the optimum-value claim above does not require uniqueness of $F^\star$ or of R.*

2.3 Unified Pipeline Pseudocode

Algorithm 1: Canonical ranking pipeline (training-free)

```
Input: Weighted digraph G = (V, E, w); wall-clock budget T; zero tolerance τ; optional min-cut
       flag; refinement budgets
Output: Score vector s : V → ℝ
t_0 ← now(); r_e ← w_e, alive_e ← true for all e ∈ E;
while true do
    if now() − t_0 > T then
        break;                                                   // may exit while cyclic
    C ← FindOneCycle(alive) with deterministic vertex/edge order;
    if C = ∅ then
        break;                                                   // acyclic
    Δ ← min_{e∈C} r_e;
    if Δ ≤ 0 then
        force-kill arg min_{e∈C} r_e
    else
        r_e ← r_e − Δ for e ∈ C;
        delete {e ∈ C : r_e ≤ τ}; if none deleted then
            force-kill arg min_{e∈C} r_e
kept ← alive;                                                    // Phase A output
acyclicFlag ← (kept is a DAG);
sort excluded arcs by descending w, stable ties by edge id;
foreach excluded arc e = (u, v) in that order do
    if now() − t_0 > T then
        break
    if v ↛* u in current kept digraph then
        kept[e] ← true
;                       // Prop. 2 is claimed only when acyclicFlag held after Phase A
if min-cut repair enabled and kept is a DAG then
    apply Alg. 2 under remaining budget
;
;                       // if Phase A exited cyclic, skip min-cut: Prop. 3 assumes a DAG
π ← TopoSort(kept); if π = None then
    π ← (1, . . . , n);                                          // identity fallback
s_v ← max{1, n − pos_π(v)};
optionally apply adjacent-swap refinement, then order-preserving ternary refinement, under
  remaining budget;
return s
```

2.4 Phase A: Local-Ratio Cycle Breaking (Prior)

Phase A follows the local-ratio reduction of Demetrescu and Finocchi (2003): maintain residuals, find a directed cycle by deterministic DFS (vertices in increasing index order; outgoing arcs in stored order), subtract the cycle's minimum residual, and delete arcs whose residual falls to at most τ. Forced-progress safeguards kill one minimum-residual arc if floating-point reduction would otherwise stall. Phase A stops when the alive subgraph is acyclic or when the wall-clock budget expires. Acyclicity is guaranteed only on the former exit.

2.5 Phase B: Exact Reachability-Aware Reinsertion (Fidelity Correction)

After Phase A, excluded arcs are scanned once in stable descending original weight. Arc (u, v) is reinserted iff there is no directed path $v \to^\star u$ in the current kept digraph. When Phase A left a DAG, this is the exact cycle-safety test of Demetrescu and Finocchi (2003); Vahidi and Koutis (2025) and Prop. 2 applies; after a premature cyclic Phase A exit the same scan is still attempted under the remaining budget, but the proposition's DAG hypotheses need not hold. A fixed-topological-position proxy (accept iff $\mathrm{pos}(u) < \mathrm{pos}(v)$ for one fixed topological order) is weaker and is retained only as an ablation baseline (INS multipass variants).

Proposition 2 (Exact add-back correctness)**.** *Assume the Phase A kept graph is a DAG. Consider a single descending-weight scan that inserts (u, v) iff $v \not\to^\star u$ in the current kept DAG.*

1. ***Acyclicity.*** *Every accepted insertion preserves acyclicity.*
2. ***Monotonicity.*** *Insertions only enlarge reachability.*
3. ***One-pass sufficiency.*** *An arc rejected when scanned cannot become admissible later in the same monotone scan.*
4. ***Single-edge inclusion minimality.*** *After a complete exact scan, no excluded arc can be restored individually without creating a cycle.*

This is inclusion-minimality under single-edge restoration, not a claim of globally minimum-weight FAS.

Proof. (1) If adding (u, v) created a cycle, the remainder of that cycle would be a $v \to^\star u$ path in the previous DAG, contradicting the acceptance test. (2) Adding arcs cannot destroy existing paths. (3) Suppose (u, v) is unsafe at its turn, so a path $v \to^\star u$ exists. Later insertions only add arcs, so that path persists and (u, v) remains unsafe. (4) Completeness of the scan means every excluded arc was rejected exactly because restoring it alone would close a cycle. □

2.6 Optional Secondary Repair: Weighted Min-Cut Exchange

Some high-weight excluded arcs remain unsafe because v reaches u. For such an arc $e = (u, v)$, let C be a minimum-capacity directed $v \to u$ cut in the retained DAG (capacities = kept-arc weights). The candidate exchange is

$$D' = (D \setminus C) \cup \{e\}.$$

Accept only if $w(C) + \mathrm{tol} < w(e)$.

Algorithm 2: Conflict-triggered weighted min-cut exchange (optional)

Input: Kept DAG D; excluded arcs; budgets on candidates/acceptances; tolerance tol
foreach *unsafe candidate $e = (u, v)$ under a fixed deterministic order* **do**
  $C \leftarrow$ directed min $v \to u$ cut in D;
  if $w(C) + \mathrm{tol} < w(e)$ **then**
    $D \leftarrow (D \setminus C) \cup \{e\}$; `// strict weighted-FAS improvement`

Proposition 3 (Min-cut exchange)**.** *Whenever an exchange is accepted under the strict rule $w(C) + \mathrm{tol} < w(e)$:*

1. *D' is acyclic;*
2. *the removed-FAS weight changes by $\Delta = w(C) - w(e) < 0$;*
3. *repeated accepted exchanges terminate after finitely many steps.*

We claim no global optimality and no approximation ratio for this operator. The guarantee is structural improvement of the weighted feedback objective; ranking metrics need not improve monotonically.

Proof. (1) Removing C destroys every $v \to u$ path, so adding (u, v) cannot close a cycle. (2) Relative to the previous feedback set, arcs in C re-enter the feedback set and e leaves it, giving $\Delta = w(C) - w(e)$; acceptance forces $\Delta < 0$. (3) There are finitely many subsets of E, and each acceptance strictly decreases a nonnegative objective bounded below by 0. □

2.7 Phase C: Scores and Optional Refinement (Prior Lineage)

From a kept DAG we compute a deterministic topological order (Kahn's algorithm with index-ordered initial queue) and assign $s_v = \max\{1, n - \mathrm{pos}(v)\}$. If the kept graph is still cyclic (e.g., Phase A timeout), the implementation falls back to the identity order $(1, \ldots, n)$; see §2.10. Optional adjacent-swap refinement may change the order under a local budget. Optional order-preserving ternary refinement adjusts magnitudes without changing the induced order, following the refinement lineage in Vahidi and Koutis (2025).

2.8 Termination, Parameters, and Determinism

Parameter	**Role**	**Notes**
T (`time_limit_sec`)	Global wall-clock budget	Canonical non-GNN: 1800 s; shared by Phases A–C; early exit may leave cycles
τ (`zero_tol`)	Residual kill threshold	Default 10^{-15}; sensitivity Z12/Z15/Z18 in §3.8
DFS / Kahn order	Deterministic traversal	Vertices $1..n$; stable edge order; C0/C1 cycle-selection in §3.8
Add-back order	Descending w, stable ties	Exact reachability test (canonical)
Legacy $P \in \{1, 2, 3\}$ (P1/P2/P3)	Fixed-topological-position passes	INS1/2/3 ablation baselines; not canonical
Min-cut budgets K20/K50/K100	Optional repair caps	Candidate attempt caps in §3.8; secondary
Refinement R0–R3	Phase C budgets	R0 = off; R1/R2/R3 scale adjacent-swap + ternary passes/time (canonical R2)
Identity fallback	Score extraction if cyclic	No nontrivial general error bound (§2.10)

Table 2: Consolidated parameters and termination behavior. Canonical OURS-Reach uses exact reachability add-back, R2 refinement (adjacent-swap then ternary), and min-cut off.

2.9 DF03 Approximation Guarantee: Idealized vs. Practical

Idealized local-ratio Phase A. If Phase A is executed under the assumptions of the published Demetrescu–Finocchi local-ratio procedure and runs to the relevant completion condition (fully reduced acyclic residual), the published λ-approximation on removed weight applies to that idealized algorithm, where λ is governed by longest-cycle length in the source analysis Demetrescu and Finocchi (2003).

Practical implementation. The implemented pipeline additionally uses finite wall-clock termination, floating-point zero tolerance, forced-progress safeguards, and a fallback ranking when topological sort fails. Therefore the original theorem does *not* automatically imply that every practical finite-budget output satisfies the same approximation guarantee.

Remark 2 (No inherited guarantee under premature termination)**.** ***We do not claim the DF03 approximation guarantee for prematurely terminated practical runs.*** *In particular, a wall-clock cutoff inside Phase A may leave the kept graph cyclic; subsequent identity-order fallback is outside the theorem's scope.*

2.10 Fallback Ordering: Unbounded Multiplicative Gap

Proposition 4 (No nontrivial multiplicative bound for identity fallback)**.** *For every* $M > 0$ *there exists a weighted digraph* G *with* $\mathrm{OPT}_{\mathrm{rank}} > 0$ *such that if* R_{fb} *denotes the identity ranking* $R_{\mathrm{fb}}(v) = v$*, then*

$$\frac{L(R_{\mathrm{fb}})}{\mathrm{OPT}_{\mathrm{rank}}} > M.$$

Proof. Fix $\varepsilon \in (0, 1)$ and an integer $n \geq 3$ large enough that $\binom{n}{2}/\varepsilon > M$. On vertex set $\{1, \ldots, n\}$, include every arc (j, i) with $j > i$ and weight 1, and add the single arc $(1, 2)$ with weight ε. Under the identity ranking $R_{\mathrm{fb}}(v) = v$, every arc (j, i) with $j > i$ is backward, so $L(R_{\mathrm{fb}}) = \binom{n}{2}$. Under the reverse ranking $R^\star(v) = n + 1 - v$, every unit-weight arc is forward and only $(1, 2)$ is backward, so $L(R^\star) = \varepsilon$ and thus $\mathrm{OPT}_{\mathrm{rank}} \leq \varepsilon$. Hence $L(R_{\mathrm{fb}})/\mathrm{OPT}_{\mathrm{rank}} \geq \binom{n}{2}/\varepsilon > M$. (The same instances show that additive regret $L(R_{\mathrm{fb}}) - \mathrm{OPT}_{\mathrm{rank}}$ is likewise unbounded as n grows with ε fixed.) □

2.11 Complexity

Let $n = |V|$ and $m = |E|$. For the audited Phase A implementation, each cycle search scans the static adjacency of all original arcs (dead arcs are skipped but not compacted), costing $O(n+m)$ per iteration, and at most m iterations occur before acyclicity in the unbudgeted case. Therefore the idealized full-completion Phase A bound is

$$O(mn + m^2),$$

which is weaker than a textbook $O(nm)$ accounting that assumes compacted alive adjacency. A wall-clock budget caps completed iterations empirically; it is an execution policy, not an average-case complexity theorem.

Exact reachability add-back sorts the m candidate arcs once ($O(m \log m)$) and, in the dense implementation used for suite graphs with $n \leq 4000$, maintains an $n \times n$ reachability matrix with an $O(n^2)$ incremental update per accepted insertion (at most m insertions), i.e. $O(m \log m + mn^2)$ in the worst case; a sparse BFS/DFS fallback is used only for larger n. Optional min-cut repair adds budgeted directed min-cut computations. Phase C is time-bounded and may use up to $O(n^2)$ pairwise structures for ratio refinement. Empirical timing is reported in §3.4–3.6.

3 Experimental Results

3.1 Experimental Protocol

Dataset suite and denominators. We use an 80-dataset manuscript benchmark suite aligned with recent ranking-from-comparisons evaluations (He et al., 2022), spanning basketball/football temporal graphs, faculty hiring networks, animal and head-to-head competition graphs, one ERO-style instance, and the near-complete Finance graph ($n = 1315$, $m \approx 1.73 \times 10^6$, density ≈ 1). Two suite members lack loadable adjacency files in the archived release (`ERO/p5K5N350eta10styleuniform`, `Halo2BetaData/HeadToHead`), so analyses that require readable graphs use an effective denominator of 78 loadable datasets ($80 - 2 = 78$). The loadable Halo family member is `Halo2BetaData`; `Halo2BetaData/HeadToHead` is a distinct suite ID without a resolvable `adj.npz` under the repository loader. An extra inventory entry `_AUTO/Basketball_temporal__1985adj` is excluded from the manuscript suite (it is a duplicate adjacency variant of Basketball 1985 and would otherwise inflate a raw repository count to 81). We never mix denominators: pairwise quality/runtime tables use method-pair common completions (typically $n = 77$ for OURS vs classical; $n = 60$ for OURS vs GNN), coverage uses the 78 loadable graphs, and family-aware macros exclude Finance from quality aggregates. Table 3 summarizes families; non-Finance ablation successes reach $n \leq 602$.

Family	#	n min–max	m min–max
Basketball (coarse+finer)	60	282–351	2,904–7,650
Football (coarse+finer)	12	20–20	107–380
Faculty (3)	3	113–206	1,204–1,787
Halo / head-to-head	2	602–602	5,010–5,010
Animal	1	21	193
ERO-style	1	(suite range)	(suite range)
Finance	1	1,315	1,729,225

Table 3: Manuscript 80-suite families (intended counts). Two missing adjacency files reduce loadable graphs to 78.

Methods. Baselines evaluated in the archived suite protocol are ten classical methods (SpringRank, SyncRank, SerialRank, BTL, DavidScore, eigenvector centrality, PageRank, RankCentrality, SVD-RS, SVD-NRS) and two GNNRank backbones (DIGRAC, ib) under fixed archived configurations (not post-hoc best-in-suite oracles). Principal Tables 4–6 report a *prespecified* principal subset of eight classical methods—SpringRank, SyncRank, SerialRank, BTL, DavidScore, PageRank, RankCentrality, SVD-NRS—plus the two GNN backbones. SVD-RS and eigenvector centrality were evaluated in the same suite export but omitted from these principal tables as near-duplicate spectral companions of SVD-NRS and PageRank, respectively (selection fixed before headline regeneration; not a post-hoc cull by effect size). An inherited GNNRank implementation bug in RankCentrality (normalized transition matrix overwritten before the eigenvector step) was corrected to the Negahban–Oh–Shah construction before regenerating that baseline's rows; other baselines were left unchanged. The *canonical* method reported in the principal quality/runtime tables is OURS-Reach: Phase A + exact reachability-aware reinsertion + Phase C refinement (adjacent-swap then order-preserving ternary; §2), *without* the optional min-cut exchange. Optional min-cut remains a secondary structural operator evaluated in ablation/empirics (§3.7, §3.9). Archived OURS-MFAS/INS3 labels correspond to duplicate fixed-topological-position configurations retained only as ablation contrasts.

Determinism, repetitions, and timing. Non-GNN methods are deterministic given fixed inputs and completed work: ranking outputs do not vary across the protocol's 10 recorded trials except through wall-clock/timeout effects. Those repetitions characterize runtime variability and timeout repeatability; they are *not* independent ranking observations. GNN methods are stochastic under training. Non-GNN wall-clock budgets use 1800 s; GNN runs use 7200 s. Archived GNN timings measure the end-to-end trained pipeline (construction + training + scoring), not inference-only latency; the exact archived GPU model ledger is unavailable, so hardware comparability is imperfect and GNN speedups are qualified accordingly.

Completion rules and statistics. Timeouts, missing data, and GNN not-applicable cases are reported separately; we do not impute NaNs as scores. Primary comparisons use pairwise common completions. We report win/tie/loss (W/T/L) counts with ties $|\Delta| \le 10^{-6}$, median paired Δ (OURS-Reach − baseline; for upset metrics, negative favors OURS-Reach), and—where precomputed—Wilcoxon signed-rank p-values with Holm adjustment and bootstrap CIs. Because many basketball seasons are temporally related, we add a family-aware sensitivity analysis (§3.3).

Canonical numerical source. Principal Tables 4–6 use OURS-Reach under the GNNRank upset metrics of He et al. (2022), paired against the archived baseline leaderboard. Structural ablation results document stage contributions and the fixed-topological-position versus reachability contrast (§3.7).

3.2 Direct Ranking Quality on Common Completions

Table 4 and Table 5 summarize OURS-Reach against principal baselines on pairwise common completions.

Baseline	n	W/T/L	med Δ	Holm p
SpringRank	77	64/0/13	−0.157	1.1×10^{-10}
DavidScore	77	75/0/2	−0.180	$< 10^{-11}$
SVD_NRS	77	76/0/1	−0.165	$< 10^{-11}$
BTL	77	76/0/1	−0.422	$< 10^{-11}$
PageRank	77	76/0/1	−0.349	$< 10^{-11}$
SyncRank	77	77/0/0	−1.008	$< 10^{-11}$
SerialRank	77	77/0/0	−1.247	$< 10^{-11}$
RankCentrality	77	71/0/6	−0.299	$< 10^{-11}$
DIGRAC	60	60/0/0	−0.316	4.9×10^{-11}
ib	60	60/0/0	−0.620	4.9×10^{-11}

Table 4: `upset_simple` (lower better) on pairwise common completions for OURS-Reach. Δ = OURS-Reach − baseline (negative favors OURS-Reach). Holm p within this table's primary family. RankCentrality uses the corrected Negahban–Oh–Shah implementation (§3.1).

Baseline	n	W/T/L	med Δ	Note
SpringRank	77	42/0/35	−0.030	competitive
DavidScore	77	74/0/3	−0.084	
SVD_NRS	77	72/0/5	−0.089	
BTL	77	3/0/74	+0.081	BTL stronger
PageRank	77	14/0/63	+0.029	PageRank stronger
SyncRank	77	47/0/30	−0.210	
SerialRank	77	47/0/30	−0.125	
RankCentrality	77	9/0/68	+0.098	RC stronger
DIGRAC	60	30/0/30	−0.039	
ib	60	30/0/30	−0.149	

Table 5: `upset_ratio` (lower better) on the same common completions for OURS-Reach. Oracle best-in-suite envelopes are not headline competitors. Corrected RankCentrality is stronger on this metric (as are BTL and PageRank).

On `upset_simple`, OURS-Reach records large W/T/L margins and significant Holm-adjusted Wilcoxon advantages against all listed classical and GNN baselines in Table 4. On `upset_naive`, the same directional pattern holds (e.g., 46/0/31 vs SpringRank; 76/0/1 vs BTL; 60/0/0 vs ib; 47/0/30 vs corrected RankCentrality). On `upset_ratio`, BTL remains clearly stronger (3/0/74), PageRank also wins more often (14/0/63), and corrected RankCentrality is stronger still (9/0/68); SpringRank is competitive (42/0/35). We do not claim universal dominance on any metric.

3.3 Family-Aware Robustness

Basketball seasons dominate per-dataset n ($\approx$ 60/77). Equal-family macros (seven included families; Finance excluded from quality macros) preserve `upset_simple` advantages for OURS-Reach vs BTL/PageRank/DavidScore/SVD_N on six of seven families. The Halo family is the main SpringRank exception; leave-one-family-out of Basketball-coarse moves the equal-family mean near zero, so SpringRank comparisons remain family-sensitive even though the pairwise Holm test is significant. The BTL `upset_ratio` disadvantage persists under equal-family scrutiny. Family-level n is too small for heavy per-family significance testing; we treat family-aware results as sensitivity, not as a replacement for pairwise tests.

3.4 Runtime and Coverage

Baseline	n	OURS faster	ties	slower	median ratio (OURS/base)
SyncRank	77	48	0	29	0.80× (∼ 1.2× faster)
SpringRank	77	15	0	62	5.7×
BTL	77	19	0	58	7.9×
RankCentrality	77	38	0	39	1.03× (comparable)
SerialRank	77	0	0	77	32×
SVD_NRS	77	0	0	77	60×
DavidScore	77	0	0	77	108×
PageRank	77	0	0	77	170×
DIGRAC	60	60	0	0	0.022× (∼ 45× faster)
ib	60	60	0	0	0.027× (∼ 37× faster)

Table 6: Paired runtime W/T/L on common completions for OURS-Reach. OURS-Reach is slower than most lightweight classical methods (comparable to SyncRank and corrected RankCentrality) and faster than archived trained-GNN end-to-end runs.

Coverage on 78 loadable datasets: OURS-Reach 77/78 (98.7%; Finance timeout); most classical methods 78/78 (100%); SyncRank 77/78 (Finance timeout); DIGRAC/ib 61/78 (78.2%, primarily not-applicable rather than timeout). A global 60-dataset subset where all methods finish exists but is GNN-applicability biased; pairwise completion remains primary. Figure 1 shows empirical OURS-Reach runtimes versus m for non-Finance graphs.

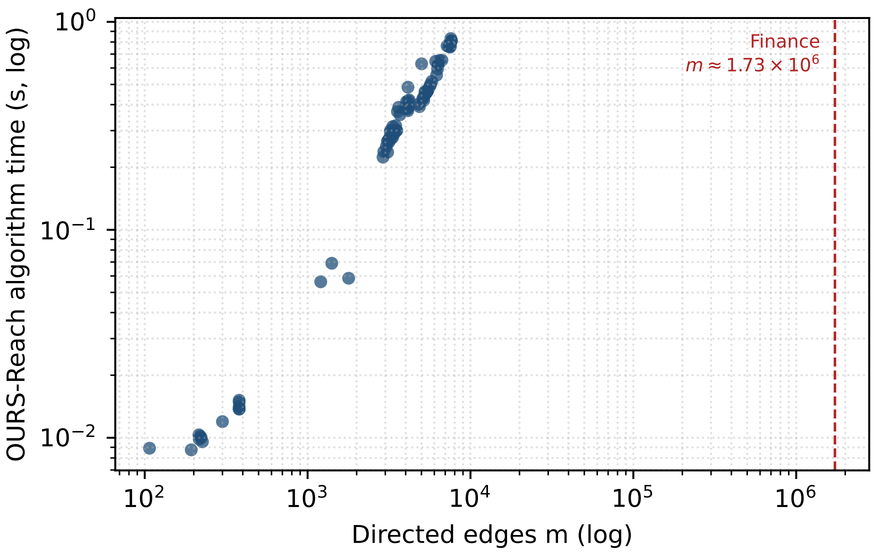

Figure 1: OURS-Reach wall time versus edge count on non-Finance graphs (log–log). Finance ($m \approx 1.73 \times 10^6$) is marked as the large-dense boundary.

3.5 Finance Stress Case

Finance is reported separately, not averaged into quality macros. Under the archived 1800 s non-GNN protocol, fixed-topological-position OURS variants and SyncRank time out while lightweight classical methods finish in < 4 s and GNN end-to-end runs finish in ~ 15–18 min. Under a tightened Phase A wall-clock budget of 600 s, Phase A-only has single-invocation algorithm wall time ≈ 600.5 s (Phase A ≈ 600.0 s hitting its budget, plus refinement ≈ 0.5 s; harness wall time, including data loading and metric computation, ≈ 612.6 s). Reachability and reachability+refinement configurations also have Phase A hit the same 600 s budget: true single-invocation algorithm wall time is ≈ 602.3 s (add-back ≈ 1.8 s and refinement ≈ 0.5 s beyond Phase A). The per-run harness timer additionally records $\approx 1214.8/1214.6$ s for these two configurations because it also executes a diagnostic Phase-A-only rerun (used only to compute the permutation-distance sensitivity statistic of §3.8) inside the same timed window; this second pass is not part of a single OURS-Reach invocation and is excluded from the algorithm wall time above. Enabling optional min-cut hits a hard wall-clock timeout at ≈ 1800.1 s without a finished ranking. This matches the $O(mn+m^2)$ Phase A discussion in §2.11: Finance is a known large-dense boundary.

3.6 Scalability (Empirical)

On non-Finance ablation graphs ($n \leq 602$), OURS-Reach single-invocation algorithm wall time is roughly 0.01–0.83 s (median ≈ 0.38 s); enabling optional min-cut adds about 3–4 s median. These timings illustrate the audited $O(mn + m^2)$ Phase A bound (§2.11) on sparse/moderate instances; the Finance boundary is in §3.5. No fitted complexity law is claimed.

3.7 Structural Ablation

We use compact stage labels A0–A6: A0 = Phase A only; A1 = fixed-topological-position add-back; A2 = exact reachability add-back; A4 = reachability+refinement (OURS-Reach); A5/A6 = optional min-cut on top of A2/A4 (Layer-1 vs full). Primary Holm family (exact CSV values):

Pair	**Metric**	n	W/T/L	med Δ	Holm p
A0→A2	upset_simple↓	77	76/0/1	-0.0166	2.4×10^{-12}
A1→A2	upset_simple↓	33	32/0/1	-0.0159	1.5×10^{-6}
A2→A5	removed weight↓	33	26/7/0	-47	8.3×10^{-6}
A4→A6	removed weight↓	77	66/11/0	-57	4.9×10^{-12}
A0→A4	upset_simple↓	77	76/0/1	-0.0169	2.4×10^{-12}

Table 7: Primary structural ablation paired tests (non-Finance). Ranking metrics need not move monotonically with the weighted FAS objective (e.g., A4→A6). A2→A5 and A4→A6 report *structural* removed-weight improvement from the min-cut operator on the retained graph; Phase C refinement does not alter kept arcs, so these FAS-weight claims do not depend on score-refinement order.

Figure 2 shows unpaired matched-support median trajectories for the legacy and canonical stage progressions (analogous to Table 8); paired tests above are authoritative. Optional min-cut configs A5/A6 are omitted from the figure because they are secondary structural operators with different support/purpose; their paired FAS-weight tests appear in Table 7.

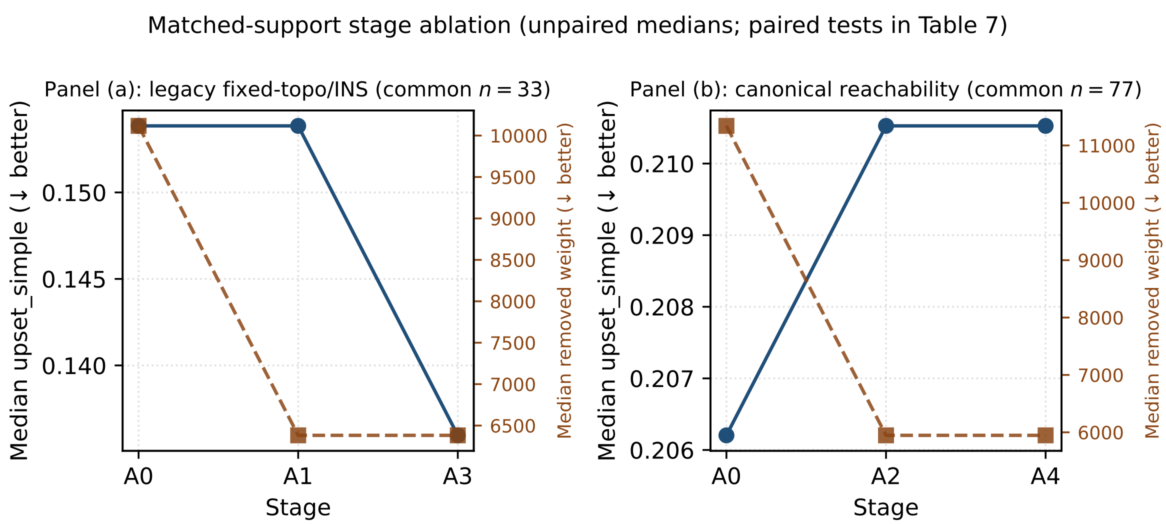

Figure 2: Matched-support unpaired medians for (a) legacy A0→A1→A3 on common $n = 33$ and (b) canonical A0→A2→A4 on common $n = 77$. Dual axes show median `upset_simple` and median removed weight within each panel. Cross-panel numerical comparisons are not intended; A5/A6 are excluded from this trajectory.

Table 8 reports `upset_simple`, `upset_naive`, `upset_ratio`, and mean single-invocation algorithm runtime across three pipeline stages, on matched dataset support. The legacy fixed-topological/INS reinsertion (A1) and its refined full pipeline (A3) were run on a smaller $n = 33$ scope than the canonical reachability stages ($n = 77$); comparing them against a Phase A row computed on the larger scope would confound the stage effect with a change in dataset composition. We therefore report two panels, each internally on one exact common-completion dataset set: Panel (a) compares Phase A only, legacy reinsertion, and the legacy refined pipeline on their shared $n = 33$ scope; Panel (b) compares Phase A only, exact reachability reinsertion, and refinement (OURS-Reach) on their shared $n = 77$ scope. Within each panel, mean `upset_simple`, `upset_naive`, and `upset_ratio` decrease monotonically from Phase A only to the full pipeline, and mean algorithm runtime increases monotonically with pipeline depth; the significant paired improvement of exact reachability add-back over Phase A only and over the legacy proxy is already quantified in Table 7, which these per-panel means complement rather than replace. Runtime is reported as `runtime_algorithm_sec` (Phase A + Phase B + Phase C of one OURS-Reach call, none of the configurations in this table use min-cut), not the harness-level `runtime_total_sec` recorded per run: for every stage past Phase A only, the per-run harness timer also executes a diagnostic Phase-A-only rerun (used solely to compute a permutation-distance sensitivity statistic, §3.8) inside the same timed window, so `runtime_total_sec` is inflated by roughly one extra Phase-A cost and is not the wall time of a single invocation.

Panel (a): legacy fixed-topological/ins progression — common $n = 33$

Stage	`upset_simple`	`upset_naive`	`upset_ratio`	Mean algorithm runtime (s)
Phase A only (A0)	0.2571	93621.2	0.3762	0.122
+ legacy fixed-topological/ins reinsertion (A1)	0.2555	93589.0	0.3760	0.127
+ legacy refinement, full pipeline (A3)	0.2521	93548.8	0.3462	0.227

Panel (b): canonical reachability progression — common $n = 77$

Stage	`upset_simple`	`upset_naive`	`upset_ratio`	Mean algorithm runtime (s)
Phase A only (A0)	0.2853	120842.9	0.3498	0.144
+ exact reachability reinsertion (A2)	0.2728	118788.6	0.3410	0.252
+ refinement, OURS-Reach (A4)	0.2724	118722.0	0.3178	0.364

Table 8: Stage-wise ablation on matched dataset support. Panel (a) reports the legacy fixed-topological/ins progression on its common-completion subset ($n = 33$); Panel (b) reports the canonical reachability-based progression on its common-completion subset ($n = 77$). Entries are means over the indicated support; mean single-invocation algorithm runtime (`runtime_algorithm_sec` = Phase A + Phase B + Phase C) is reported alongside the three upset losses (primary runtime comparisons elsewhere use the median, as in Table 6). The 33 datasets in Panel (a) are a subset of those in Panel (b), so cross-panel numerical comparisons are not intended; paired significance is reported in Table 7.

3.8 Parameter Sensitivity (Compact)

On Layer 1 ($n = 33$): `zero_tol` $\in \{10^{-12}, 10^{-15}, 10^{-18}\}$ is STABLE (almost all ties vs default). Cycle-selection changes are statistically detectable on the reachability+refinement path but small ($|\text{med}\,\Delta| \approx 10^{-3}$). Refinement: R0 (off) is worse; R1≈R2 and R3≈R2 (saturation). Fixed-topological multipass add-back: P1 restores many edges; P2/P3 add little ranking quality. Min-cut budgets: K20→K50 modest; K50→K100 nearly flat. The main sensitivity trends relevant to the conclusions are summarized here.

3.9 Min-Cut Empirical Regime

The optional min-cut exchange is a *secondary* structural contribution: it is not part of OURS-Reach (Tables 4–6). In the ablation harness, A5 applies min-cut after reachability without Phase C; A6 runs Phase C refinement before the exchange and then extracts post-exchange topological scores for upset reporting (so A6 upset deltas are not "A4 scores plus min-cut"). Phase C does not modify the retained arc set, so the A2→A5 and A4→A6 *removed-weight* improvements in Table 7 remain valid structural evidence for Proposition 3. Under the predefined broad characterization (39 available of 40 datasets), the operator of §2.6 is active on 28/39 graphs (4/7 families; 280 accepted exchanges). On active graphs, `upset_simple`/`upset_naive` never worsened; `upset_ratio` worsened on 7/28 (Basketball only; small magnitude), confirming regime-specific rather than universal ranking benefit. We therefore claim monotone weighted-FAS improvement where exchanges are accepted, not uniform ranking dominance from min-cut.

4 Limitations

Dense near-complete graphs remain difficult: Finance is the explicit boundary case documented in §3.5. Finite-budget practical outputs need not inherit the idealized Demetrescu–Finocchi guarantee (§2.9), and identity-order fallback has no nontrivial general multiplicative error bound (Prop. 4). The optional min-cut repair improves the weighted feedback objective but not every upset metric, and its benefit is regime-specific (§3.9). Relative to most lightweight classical estimators OURS-REACH is slower (§3.4); GNN speedups refer to archived end-to-end training+evaluation under imperfect hardware comparability (§3.1). Family-aware analysis shows that SpringRank comparisons remain family-sensitive, and BTL remains stronger on `upset_ratio` under equal-family scrutiny (§3.3). Corrected RankCentrality is stronger than OURS-REACH on pairwise `upset_ratio` (Table 5); family-aware RankCentrality macros were not recomputed after the baseline correction and are not used as evidence. Protocol repetitions for deterministic non-GNN methods characterize runtime/timeout variability rather than ranking uncertainty.

5 Conclusion

We addressed ranking from weighted pairwise comparisons with a training-free structural pipeline: local-ratio cycle breaking, exact reachability-aware reinsertion, optional weighted min-cut exchange, and score extraction with optional Phase C refinement (adjacent-swap then order-preserving ternary). The algorithmic core follows Demetrescu–Finocchi and Vahidi and Koutis (2025); the contributions of this paper are formal ranking–MWFAS clarification and guarantee boundaries, restoration of exact cycle-safe add-back relative to a weaker fixed-topological-position rule, a secondary monotone min-cut repair, and a broader timeout-aware empirical study.

On common completions, OURS-REACH is strong on simple/naive upset metrics against the listed classical and GNN baselines, remains weaker than BTL and corrected RankCentrality on upset_ratio, and is family-sensitive versus SpringRank despite a significant pairwise Holm test on upset_simple. It is slower than most lightweight classical methods (comparable to SyncRank and corrected RankCentrality), substantially faster than the evaluated trained GNN pipelines under the archived end-to-end protocol, and encounters a large-dense limit on Finance.

6 Future Work

Theoretically, denser-graph cycle breaking, tighter analyses of time-bounded structural heuristics, and approximation results on restricted graph classes remain open. Engineering directions include SCC-aware or parallel cycle processing, more efficient reachability maintenance, and better min-cut candidate selection under budgets. Application settings such as preference aggregation, multi-source sensor comparisons, and industrial ranking are natural targets for the training-free pipeline but are not evaluated here. Deterministic uncertainty quantification over multiple valid topological orders is another concrete direction.

Acknowledgments

The author would like to express sincere gratitude to his family for their continued support and encouragement, and to Professor Ioannis Koutis for his guidance and support throughout this research. The author also thanks Anders Borum for generously providing complimentary lifetime access to Secure ShellFish, which supported the remote computing workflow used during this research.

Declaration of Generative AI and AI-assisted Technologies

During the preparation and revision of this work, the author used ChatGPT (OpenAI), Claude (Anthropic), Cursor, and Perplexity AI to assist with manuscript organization and language refinement, literature and consistency checks, coding and analysis support, and preparation of responses to reviewer comments. All AI-assisted outputs, code, analyses, interpretations, citations, and manuscript changes were independently reviewed and verified by the author, who takes full responsibility for the scientific content and final manuscript.

Statements and Declarations

Funding The author received no financial support for the research, authorship, and/or publication of this article.

Competing Interests The author declares that he has no competing interests.

Ethics Approval Not applicable. This article does not report studies involving human participants or animals.

Consent to Participate Not applicable.

Consent for Publication Not applicable.

Availability of Data and Materials The datasets used in the experiments originate from prior benchmarks cited in the manuscript (e.g., He et al. (2022)). For reproducibility, the processed suite graphs used in this study are provided in the public repository linked below, primarily as sparse adjacency matrices in `.npz` form (with some families also retaining text edge-list sources). Two intended suite members lack loadable adjacency files in the archived release (`ERO/p5K5N350eta10styleuniform`, `Halo2BetaData/HeadToHead`), so loadable analyses use the 78-graph denominator stated in §3.1.

Code Availability All source code, processed data, and scripts required to reproduce the experiments are publicly available at `https://github.com/SoroushVahidi/ranking-by-feedback-arc-set`. The repository contains the implementation of the proposed method, the benchmark drivers, and the materials needed to reproduce the reported results. Implementations of the non-proposed baselines and the GNNRank training and evaluation pipeline are based on the public GNNRank codebase of He et al. (2022), with the corresponding adaptations documented in the repository.

Author Contributions S. Vahidi is the sole author. He conceived the study, developed the algorithms, performed the experiments, and wrote the manuscript.

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